

References: Read Reactor Theory Manual Chapter 1: §1.5**Reference the following content on the class Moodle page:**

- *Lecture Module #6 – Lecture Slides*
- *Lab Module #4 – 1/M Approach to Criticality PDF & Data File*

1. Sub-Critical Multiplication: Mark the Following Statements **TRUE** or **FALSE** (Assume $K_{eff} < 1$ for all cases):

- _____ a) Equal changes in K_{eff} do NOT result in equal changes in equilibrium neutron population N_{eq} .
- _____ b) As K_{eff} approaches 1.0, more generations are required to reach equilibrium; therefore, less time is required to reach equilibrium.
- _____ c) If K_{eff} is increased halfway to 1.0 from its initial value, the neutron count rate will decrease by half.
- _____ d) As K_{eff} approaches 1.0, the # of neutrons lost from the fission chain decreases (i.e. $N_i(1-K_{eff})$ decreases)

2. **Do Chapter #1 Problem #2 on page 25 of Reactor Theory Manual (use a separate piece of paper please):** Note: $K_{eff} = 1/(1-\rho)$; $CR_2 = CR_1 \times (1-K_1)/(1-K_2)$. Calculate changes in K_{eff} from step additions in reactivity and then each corresponding equilibrium count rate (CR). Sketch NFM trace.

Continued on next page....

1/M Lab Questions - Fill in the blanks:

REF: Lab #4 – 1/M Approach to Criticality PDF & Excel Data Plot (from lab)

1/M Startup - Fill in the blanks:

1. During a 1/M startup, do the following values increase ($\uparrow$), decrease ($\downarrow$) or stay the same ($\leftrightarrow$)?:
S ____ N_f ____ CR_o ____ CR_i ____
2. During a 1/M startup, what limits are approached for the following values? (0, 1, or ∞):
 $K_{eff} \rightarrow$ ____ $M \rightarrow$ ____ $1/M \rightarrow$ ____
3. Which detector in the reactor core is the Neutron Flux Monitor (NFM) connected to?
4. What are the two conditions used (*i.e. whichever happens first*) when determining how far to withdraw the control rods for each 1/M step?
5. How many times in lab did we determine how far to withdrawal the rods by doubling the count rate?
6. Approximately what critical gang rod height (in inches) did our 1/M plot converge to?
7. Apart from the first rod pull, was our plotted 1/M approach to criticality conservative or non-conservative?